

Journal of Financial Crises

Volume 2 | Issue 3

2020

Term Securities Lending Facility (TSLF) (U.S. GFC)

Manuel Leon Hoyos

Yale School of Management

Follow this and additional works at: <https://elischolar.library.yale.edu/journal-of-financial-crises>



Part of the [Finance Commons](#), and the [Macroeconomics Commons](#)

Recommended Citation

Leon Hoyos, Manuel (2020) "Term Securities Lending Facility (TSLF) (U.S. GFC)," *Journal of Financial Crises*: Vol. 2 : Iss. 3, 202-228.

DOI: <https://doi.org/10.17132/2693-3179.1106>

Available at: <https://elischolar.library.yale.edu/journal-of-financial-crises/vol2/iss3/8>

This Case Study is brought to you for free and open access by the Journal of Financial Crises and EliScholar – A Digital Platform for Scholarly Publishing at Yale. For more information, please contact journalfinancialcrises@yale.edu.

READERS TAKE NOTICE:

As of April 2024, the YPFS Resource Library's site domain has changed to <https://elischolar.library.yale.edu/ypfs-financial-crisis-resource-library/>.

Please be aware that upon clicking any of the URL references to the former Resource Library domains, either <https://ypfs.som.yale.edu> or <https://ypfsresourcelibrary.blob.core.windows.net/>, in the "References/Key Program Documents" section of a case study, readers will encounter a "page not found" error.

Readers can still retrieve a given resource cited within this case study on the [new site](#) by searching [here](#) for the title cited.

Term Securities Lending Facility (TSLF) (U.S. GFC)¹

Manuel León Hoyos^{2,3}

Yale Program on Financial Stability Case Study

March 20, 2019; Revised: October 10, 2020

Abstract

The 2007–09 financial crisis reached a critical stage in March 2008. Amid falling house prices and downgrades of mortgage-related securities, financial markets became severely disrupted. The Federal Reserve—the US central bank—became increasingly concerned about the inability of the 20 primary dealers, including the five largest US investment banks, to fund themselves in short-term funding markets, such as the repurchase agreement market, then estimated at \$10 trillion. In response, the Fed created several emergency lending facilities to restore market liquidity that required the Fed to invoke Section 13(3) of the Federal Reserve Act. The Term Securities Lending Facility authorized the Federal Reserve Bank of New York to lend to primary dealers up to \$200 billion of highly liquid US Treasuries against collateral that was particularly illiquid at the time. Eligible collateral initially included triple-A private-label mortgage-backed securities but was later broadened. In July 2008, an additional \$50 billion was allocated for a TSLF Options Program. The TSLF operated between March 27, 2008, and February 1, 2010. Usage peaked at \$236 billion in October 2008. Overall, 18 of the 20 primary dealers participated and the Fed collected \$781 million in fees.

Keywords: Federal Reserve, central bank, financial crisis, lending facilities, lender of last resort, market liquidity

¹ This case study is part of Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering the responses to the global financial crisis that pertain to market liquidity programs.

Cases are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/>.

² Manuel León Hoyos – Research Associate, YPFS, Yale School of Management.

³ Special thanks to Julia Arnous for research contributions.

Term Securities Lending Facility (TSLF)

At a Glance

In March 2008, financial markets became severely disrupted amid falling house prices and downgrades of mortgage-related securities. Particularly distressed was the repurchase agreement (repo) market, then estimated at \$10 trillion. The Federal Reserve (the Fed)—the US central bank—concerned about the inability of primary dealers to obtain funds, invoked the emergency powers of Section 13(3) of the Federal Reserve Act (FRA), last used during the Great Depression, to act as lender of last resort to nondepository institutions. The Term Securities Lending Facility (TSLF) was the first of many emergency lending facilities during the crisis.

The TSLF was used by the Federal Reserve Bank of New York (FRBNY) to lend to its 20 primary dealers, including the largest five US investment banks and the US securities arms of foreign financial institutions, up to \$200 billion of highly liquid US Treasuries against collateral that was relatively illiquid at the time. In view of the great demand for US Treasuries as a “safe haven,” the TSLF was intended to restore market liquidity, particularly for the repo market.

The list of eligible collateral initially included triple-A private-label mortgage-backed securities (MBS) but was later expanded. In July 2008, an additional \$50 billion was allocated for a TSLF Options Program (TOP). In September 2008, a day prior to the collapse of Lehman Brothers—the largest bankruptcy in US history—the Fed greatly expanded TSLF eligible collateral to include any investment-grade securities (BBB- or higher). TSLF utilization intensified, reaching its peak of \$236 billion in October 2008. The TSLF operated until February 1, 2010. Overall, 18 primary dealers participated and the Fed collected \$781 million in fees.

Summary Evaluation

Since the TSLF was one of many emergency lending facilities deployed during the 2007–09 financial crisis, it is hard to evaluate its direct impact on financial markets generally. The effect of the TSLF announcement on market participants was mixed. While Fed Chairman Bernanke (2015) thought it calmed the markets, US Treasury Secretary Paulson (2010) thought that “the opposite happened.” Fleming, Hrungr, and Keane (2010) and Hrungr and Seligman (2011) argued that the TSLF contributed to the reduction of stress on repo markets. Recently, Carlson & Macchiavelli (2018) conducted further analysis of the TSLF.

Summary of Key Terms	
Purpose:	To promote liquidity in the financing markets for Treasury and other collateral and thus to foster the functioning of financial markets more generally
Announcement Date	March 11, 2008
Operational Date	March 27, 2008
End of Issuance Window	February 1, 2010
Legal Authority	Sections 13(3) and 14 of the Federal Reserve Act
Peak Utilization	\$236 billion on October 1, 2008
Participants	Primary dealers of the Federal Reserve Bank of New York
Administrator	Federal Reserve Bank of New York

Term Securities Lending Facility: United States Context	
GDP (SAAR, Nominal GDP in LCU converted to USD)	\$14,681.5 billion in 2007 \$14,559.5 billion in 2008 <i>Source: Bloomberg</i>
GDP per capita (SAAR, Nominal GDP in LCU converted to USD)	\$47,976 in 2007 \$48,383 in 2008 <i>Source: Bloomberg</i>
Sovereign credit rating (5-year senior debt)	As of Q4, 2007: Fitch: AAA Moody's: Aaa S&P: AAA As of Q4, 2008: Fitch: AAA Moody's: Aaa S&P: AAA <i>Source: Bloomberg</i>

Size of banking system	\$9,231.7 billion in total assets in 2007 \$9,938.3 billion in total assets in 2008 <i>Source: Bloomberg</i>
Size of banking system as a percentage of GDP	62.9% in 2007 68.3% in 2008 <i>Source: Bloomberg</i>
Size of banking system as a percentage of financial system	Banking system assets equal to 29.0% of financial system in 2007 Banking system assets equal to 30.5% of financial system in 2008 <i>Source: World Bank Global Financial Development Database</i>
5-bank concentration of banking system	43.9% of total banking assets in 2007 44.9% of total banking assets in 2008 <i>Source: World Bank Global Financial Development Database</i>
Foreign involvement in banking system	22% of total banking assets in 2007 18% of total banking assets in 2008 <i>Source: World Bank Global Financial Development Database</i>
Government ownership of banking system	0% of banks owned by the state in 2008 <i>Source: World Bank, Bank Regulation and Supervision Survey</i>

Existence of deposit insurance	100% insurance on deposits up to \$100,000 for 2007 100% insurance on deposits up to \$250,000 for 2008 <i>Source: Federal Deposit Insurance Corporation</i>
---------------------------------------	---

I. Overview

Background

The financial crisis that started in the second half of 2007 entered a critical stage in March 2008. Financial markets became severely disrupted and credit was scarce and expensive. Amid falling house prices and downgrades of mortgage-related securities, lenders limited their exposure to only the safest securities. Bear Stearns Companies, the fifth-largest investment bank with \$400 billion in assets, teetered on the brink of collapse (GAO 2011; Geithner 2014).

Primary dealers⁴ such as Bear Stearns were particularly vulnerable as they relied on the repurchase agreement (repo) market for funding (Gorton and Metrick 2012). A repo transaction is basically a short-term loan in which a firm sells a security to another firm with the agreement to buy it at a later date for a slightly higher predetermined price. These transactions occur on a very short-term basis, typically overnight.

By March 2008, the repo market, estimated at \$10 trillion at the time, had become illiquid (Gorton and Metrick 2010). Primary dealers struggled to obtain financing, as they could no longer sell a large portion of their securities (Geithner 2014). Rates for overnight borrowing of instruments such as Treasury securities, agency debt securities, and agency mortgage-backed securities (MBS) rose from an average of less than 10 basis points (bps) to more than 60 bps (Fleming, Hrungr, and Keane 2010). The Federal Reserve (the Fed)—the US central bank—became concerned about market liquidity and the functioning of financial markets (Federal Reserve 2008a).

On March 7, 2008, the Fed introduced the Single-Tranche Term Repurchase Agreements program,⁵ which allocated up to \$100 billion to conduct term (28-day) repurchase agreements with primary dealers. This program excluded private-label mortgage-related securities and accepted only high-quality collateral, eligible in the Fed's regular open market operations (OMOs) in which the Federal Reserve Bank of New York (FRBNY) trades overnight securities with its primary dealers. The Fed, through the FRBNY, crafted a new and larger program, the Term Securities Lending Facility (TSLF), intended to support primary dealers. In view of the great demand for US Treasury securities as a "safe haven," and to reduce the need for a fire sale of illiquid assets, the FRBNY would lend highly liquid US Treasury securities at a 28-day term—much longer than the typical overnight term—and against a broader range of collateral (Federal Reserve 2008a; Geithner 2014). Throughout the crisis, the Fed created multiple emergency lending facilities intended to restore market liquidity, such as the TSLF.^{6,7}

⁴ Primary dealers are about 20 securities firms that the Federal Reserve Bank of New York (FRBNY) has designated to trade US government securities on a regular basis. Most primary dealers are affiliated with banks. However, this doesn't give them access to the Federal Reserve's discount window. The discount window is available only to legal entities that are depository institutions. US securities firms had limited ability to source liquidity from their bank affiliates during the crisis. For that reason, the Fed believed it needed to create programs like Term Securities Lending Facility (TSLF) to promote market liquidity by lending directly to securities firms.

⁵ For information about the Single-Tranche Term Repurchase Agreements program, see: https://www.federalreserve.gov/monetarypolicy/bst_tranche.htm.

⁶ For a crisis timeline and info on Fed's actions, see <https://www.stlouisfed.org/financial-crisis/full-timeline>.

⁷ For TSLF information, see Federal Reserve 2018 and FRBNY 2018a.

Program Description

On March 10, 2008, a week prior to a scheduled regular meeting of the Federal Open Market Committee (FOMC), Fed Chairman Bernanke convoked an emergency conference call. Financial market disruptions rushed the Fed to speed up the creation of emergency lending facilities. That same day, with a unanimous vote of the five sitting Board Governors of the Federal Reserve System and a 9–0 vote of the FOMC, the TSLF was approved. The Fed invoked the emergency powers of Section 13(3) of the Federal Reserve Act (FRA) that permitted it to lend to nondepository institutions “in unusual and exigent circumstances” (Federal Reserve 2009). The TSLF was announced a day later, on March 11, 2008, although without any reference to Section 13(3) or emergency authority. The Fed “worried that trumpeting the invocation of emergency powers last used in the Depression would deepen the panic” (Bernanke 2015).

Under the TSLF, the FRBNY was able to lend primary dealers as much as \$200 billion in highly liquid US Treasury securities such as Treasury bills, notes, bonds, and inflation-indexed securities. The facility took advantage of existing infrastructure and had some similarities in design to the FRBNY’s open market operations. The FRBNY pulled out US Treasury securities for the operation of this facility from the System Open Market Account (SOMA) (Federal Reserve 2009). However, it took two weeks for the facility to become operational. In the meantime, JPMorgan Chase & Company acquired Bear Stearns for \$2 per share with emergency assistance from the Fed.⁸

The FRBNY offered to lend US Treasury securities at a 28-day term (in some cases adjusted for holidays) against a broader range of collateral. Initially, the list of eligible collateral included illiquid collateral at the time, such as triple-A private-label residential MBS and commercial MBS, as well as agency collateralized mortgage obligations (CMOs) (FRBNY 2008a). By awarding loans through auctions, the Fed intended to encourage broad participation and avoid any stigma in using this facility. The TSLF consisted of weekly auctions (Thursdays at 2 p.m. ET) in which bids represented the fee dealers intended to pay to loan the offered US Treasury securities. At the end of each auction, the FRBNY awarded loans at a uniform fee, based on the “stop-out rate,” which was the lowest accepted bid.

TSLF auctions started on March 27, 2008, and were classified into two categories: Schedule 1 and Schedule 2. While Schedule 1 auctions accepted all collateral eligible in the FRBNY’s open market operations, Schedule 2 auctions offered a larger amount of US Treasury securities, accepted a much broader and less liquid range of collateral, and required a higher minimum bid. The two different schedules were intended “to better calibrate the interest rate on TSLF loans to the level of risk associated with the collateral” (GAO 2011).

Section 14 of the Federal Reserve Act already allowed the FRBNY to trade securities eligible for Schedule 1 auctions for purposes of OMOs. However, the TSLF required FRA Section 13(3) approval because, in expanding the collateral for Schedule 2 auctions to include types of securities that were not authorized in the FRBNY’s open market operations, it was lending to nondepository institutions (the primary dealers) for the purpose of acting as their lender of last resort.

The Fed used the clearing services of the existing clearing banks in the triparty repo market, JPMorgan Chase and Bank of New York Mellon Corporation. The clearing banks acted as

⁸ The price settled at \$10 per share. The Fed facilitated a \$30 billion emergency loan to JPMorgan Chase. Between March 10 and March 13, Bear Stearns had experienced a depletion of its cash reserves from \$18 billion to \$2 billion.

intermediaries and provided the daily services of custody and valuation of collateral. The services of the clearing banks came at no charge to the Federal Reserve or borrowers (Federal Reserve 2009).

Participation in the TSLF was voluntary and undisclosed. In July 2010, however, the US Congress passed the Dodd-Frank Wall Street Reform and Consumer Protection Act (Dodd-Frank), which requires the Fed to disclose information about OMOs and discount window borrowers, two years after transactions occur. For emergency lending facilities under Section 13(3) authority, the Fed is required to disclose information of borrowers a year after the facility ends (Federal Reserve 2010).⁹

The day before a TSLF auction, the FRBNY announced the amount of US Treasury securities offered, which ranged between \$25 billion and \$75 billion. TSLF auctions lasted for 30 minutes, and the results on awards and stop-out rates were communicated shortly thereafter (Federal Reserve 2008a).

Schedule 1 auctions imposed a minimum bid of 10 bps, while Schedule 2 required a higher minimum of 25 bps. There was no cost to place a bid. Dealers could place up to two bids, with a minimum amount of \$10 million and a maximum of 20% of the total offered amount. In case of two bids, they could only be in increments of \$10 million. The FRBNY awarded loans in full for bids above the stop-out rate and on a pro rata basis for bids at the stop-out rate and held the right to refuse any bid at its own discretion. Dealers were not allowed to terminate a loan early (FRBNY 2009).

The FRBNY imposed margins (haircuts)¹⁰ on all collateral. The TSLF terms and conditions stipulated the daily revaluation of collateral by the clearing banks to make sure the specified margins were applied. Every day, the clearing banks conducted the valuation of collateral. In case the value of the collateral decreased, the FRBNY could ask for substitutions. On the other hand, primary dealers could also substitute collateral for other eligible collateral if needed. To value the pledged collateral, clearing banks used the lowest price available in their valuation systems. This reduced the risk for the Fed in case of bankruptcy of a borrowing primary dealer. The loans were recourse. That is, in case of default, the Fed could come after the primary dealer's assets to claim the difference between the liquidation of the pledged collateral and the value of the loan (GAO 2011)

With the escalating strains on financial markets, on July 24, 2008, the FOMC arranged a conference call and voted to extend the TSLF until January 30, 2009. The FOMC also approved the TSLF Options Program (TOP) as an extension to the TSLF.¹¹ The TOP was announced on July 30, 2008. It required administrative changes pursuant to Schedule 2 auctions and not another Section 13(3) authorization (Federal Reserve 2009). The new program authorized the FRBNY to offer through auctions the option to borrow US Treasury securities for a seven-day term, two to three weeks after the auction date and at a fixed rate. The TOP intended to provide primary dealers some relief during periods of "heightened collateral market pressure, such as quarter-end dates" (Federal Reserve 2018). TOP auctions started in late August 2008 and required a minimum bid of 1 bp; all Schedule 2 collateral was eligible. The facility was limited to \$50 billion in US Treasury securities, on top of the

⁹ For the Fed's disclosures, see https://www.federalreserve.gov/newsevents/reform_quarterly_transaction.htm.

¹⁰ A margin or haircut requires the pledged collateral to be of greater value than the securities borrowed.

¹¹ Only Governor Plosser dissented. He considered "the net benefit of the TSLF options as being insufficient to justify adding them to the support already being provided to market liquidity." See <https://www.federalreserve.gov/boarddocs/rptcongress/annual08/pdf/AR08.pdf>.

\$200 billion of Schedule 1 and Schedule 2 auctions. The day before each TOP auction, the FRBNY announced the terms and conditions, including the fixed rate of the loans.

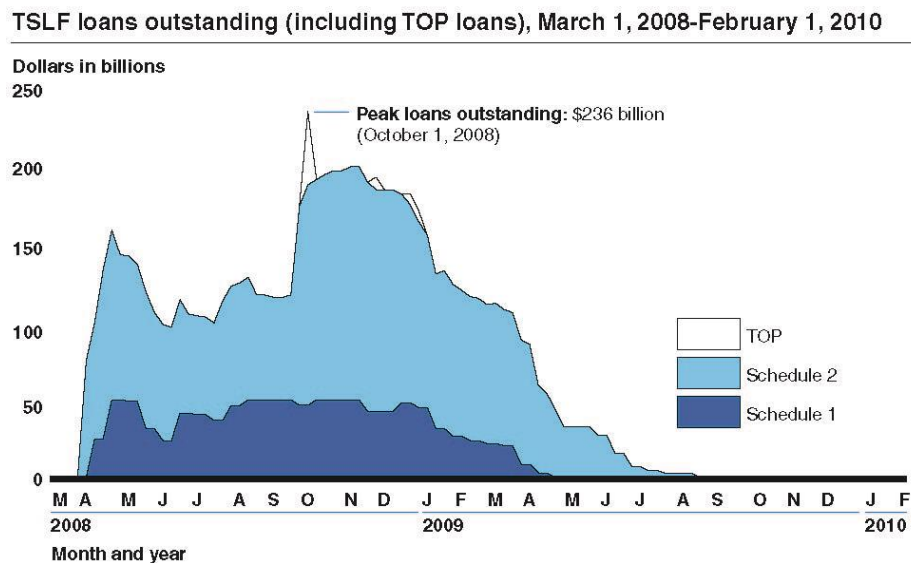
On September 14, 2008, a day prior to the failure of Lehman Brothers—the fourth largest investment bank and largest bankruptcy in US history—the Fed took another expansionary step.¹² With financial markets in disarray, TSLF Schedule 2 eligible collateral was expanded to include all investment-grade debt securities (BBB- or higher) and auctions increased from \$125 billion to \$150 billion per month. Schedule 2 auctions started to run weekly, instead of biweekly, until the end of April 2009, when they returned to a biweekly basis. In the January 2009 FOMC meeting, the TSLF was extended until October 2009 (Federal Reserve 2009).

By June 2009, financial markets showed significant signs of improvement. Effective July 1, 2009, the Fed suspended TSLF Schedule 1 and TOP auctions. The TSLF size was reduced to \$75 billion for Schedule 2 auctions, which decreased from biweekly to once a month. After July 16, 2009, auctions received no participation. The last auction took place on January 7, 2010. While the TSLF was in operation, the Fed published information on the total amount of propositions, awards, stop-out rate, and bid-to-cover ratio of each auction, but it did not disclose the identity of the participant primary dealers or the bid propositions (FRBNY 2018a).

Outcomes

The TSLF operated between March 27, 2008, and February 1, 2010. Overall, 18 of the 20 primary dealers participated in the TSLF and 11 participated in the TOP as well. All TSLF loans were paid in full and with interest. The Fed collected a total of \$781 million in fees (Federal Reserve 2010). The TSLF reached its peak utilization of \$236 billion on October 1, 2008. See figures 1 and 2 below.

Figure 1: TSLF Utilization



Source: GAO 2011.

¹² Lehman Brothers had been an active participant of the TSLF up until the September 11, 2008, auction. The investment bank received 18 TSLF loans. Its total borrowing peak was \$19 billion.

Figure 2: List of Primary Dealers of the Federal Reserve Bank of New York

	Primary Dealer	Schedule 1	Schedule 2	TOP	TSLF Loans	Borrowing Peak (dollars in millions)
1	Credit Suisse Securities (USA) LLC	11	42	1	53	\$38,510
2	Morgan Stanley & Co. Incorporated	6	28	3	34	\$36,000
3	Goldman Sachs & Co.	15	38	2	53	\$34,500
4	Deutsche Bank Securities Inc.	20	32	1	52	\$34,284
5	Citigroup Global Markets Inc.	20	45	2	65	\$34,100
6	RBS Securities Inc.	14	44	1	58	\$32,200
7	Barclays Capital Inc.	21	44	1	65	\$26,200
8	UBS Securities LLC.	4	17	0	21	\$23,823
9	Merrill Lynch Government Securities Inc.	5	34	1	39	\$21,777
10	Lehman Brothers Inc.	5	13	0	18	\$19,000
11	Banc of America Securities LLC	8	15	1	23	\$17,203
12	J.P. Morgan Securities LLC	7	16	2	23	\$13,000
13	BNP Paribas Securities Corp.	9	12	2	21	\$11,500
14	Countrywide Securities Corporation	5	5	0	10	\$3,600
15	Bear Stearns & Co., Inc.	0	2	0	2	\$2,000
16	Dresdner Kleinwort Securities LLC	2	0	0	2	\$850
17	Cantor Fitzgerald & Co.	4	5	0	9	\$700
18	HSBC Securities (USA) Inc.	0	11	0	11	\$500
19	Daiwa Securities America Inc.	-	-	-	0	
20	Mizuho Securities USA Inc.	-	-	-	0	
21	Jefferies & Company, Inc. (*)	-	-	-	0	
22	RBC Capital Markets Corporation (*)	-	-	-	0	
	TOTAL	156	403	17	559	

**The institution became a primary dealer in June or July 2009.*

Source: Author's analysis of Federal Reserve data.

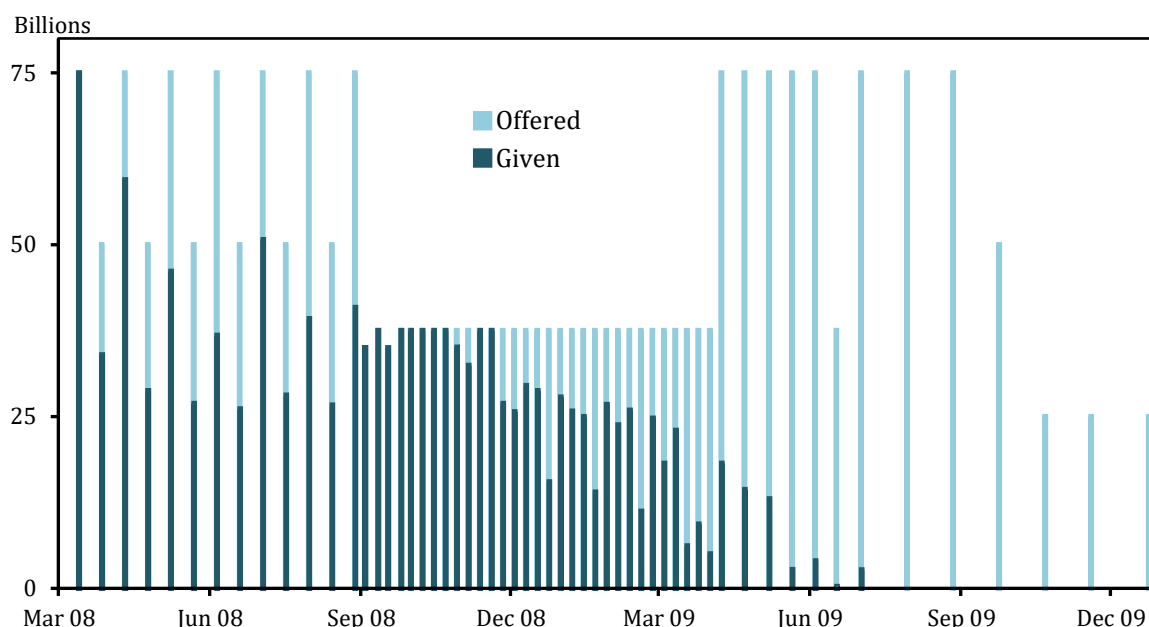
The FRBNY held a total of 97 TSLF auctions: 58 Schedule 2, 33 Schedule 1, and six TOP. Overall, the FRBNY awarded a total of 559 TSLF loans; 156 Schedule 1 and 403 Schedule 2. Of the 403 Schedule 2 collateral loans, 17 were TOP. Schedule 2 auctions received the most demand. The highest stop-out rates occurred in the Schedule 2 auctions of September 17, October 9, and October 15, at 300 bps, 305 bps, and 322 bps, respectively.¹³

TSLF auctions started on March 27, 2008, and ran weekly, alternating between Schedule 1 and Schedule 2, until September 11, 2008. In these first months of operation, the Fed offered \$175 billion of the available \$200 billion: \$50 billion for Schedule 1 auctions and \$125 billion for Schedule 2 auctions. Schedule 1 auctions offered \$25 billion on each auction, while

¹³ For TSLF data and statistics, see Federal Reserve 2018 and FRBNY 2018b.

Schedule 2 auctions alternated offering amounts between \$50 billion and \$75 billion. See figure 3 below.

Figure 3: Utilization of TSLF Schedule 2



Note: Each bar represents a Schedule 2 auction. In total, there were 58 between March 27, 2008, and January 7, 2010.

Source: Author's analysis of FRBNY data.

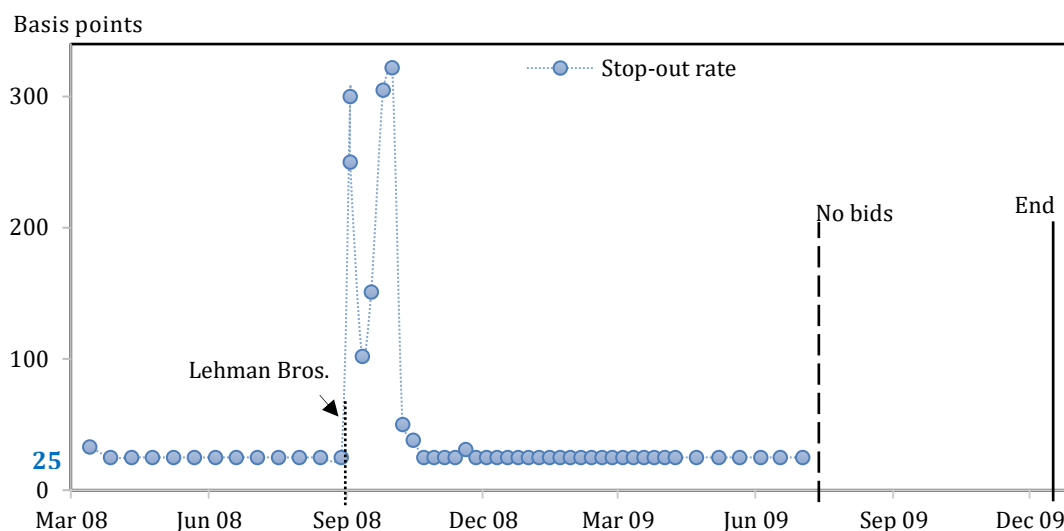
The initial TSLF auction on March 27, 2008, was Schedule 2. It offered \$75 billion in US Treasury securities and was oversubscribed. The FRBNY awarded loans to 15 dealers at a stop-out rate of 33 bps. Schedule 2 auctions offered a significantly larger amount of US Treasury securities compared to Schedule 1, which offered \$25 billion in all its TSLF auctions. With the exception of the first Schedule 2 auction of March 27, 2008, all Schedule 2 auctions until September 11, 2008, were undersubscribed. Following the collapse of Lehman Brothers on September 15, 2008, Schedule 2 auctions ran weekly, instead of biweekly, and the Fed increased its offerings from \$125 billion to \$150 billion per month. From September 17, 2008, until April 15, 2009, all Schedule 2 weekly auctions offered \$37.5 billion (only three \$35 billion).

On September 17, 2008, \$70 billion in US Treasury securities were split into two auctions of \$35 billion each. The first auction introduced a relatively shorter term of 14 days (term offered only once), while the one later the same day offered the regular 28-day term. Both of the September 17, 2008, auctions were fully subscribed at stop-out rates of 250 bps and 300 bps, respectively. All Schedule 2 auctions from September 17, 2008, until November 5, 2008, were oversubscribed.

Starting on April 22, 2009, Schedule 2 auctions returned to a biweekly basis and offered \$75 billion each. From December 10, 2008, until July 16, 2009, auctions were

undersubscribed. Thereafter, participation stopped completely and Schedule 2 auctions started to phase out, eventually ending on January 7, 2010.

Figure 4: Stop-Out Rates for TSLF Schedule 2

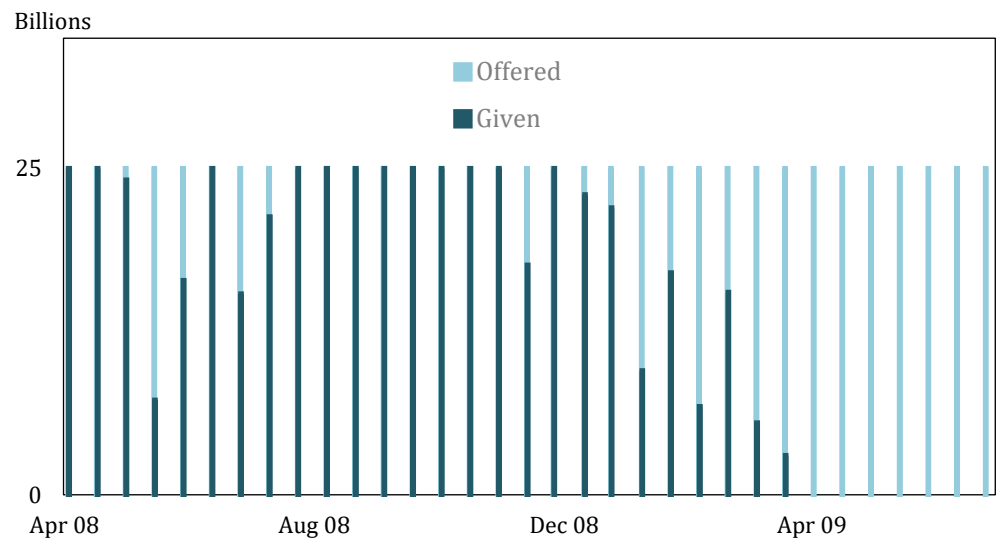


Note: The minimum bid rate was 25 bps. No bids after July 16, 2009. The last auction was held on January 7, 2010.

Source: Author's analysis of FRBNY data.

There were 33 biweekly Schedule 1 auctions between April 3, 2008, and June 25, 2009. Each offered \$25 billion. After March 19, 2009, there was no participation. From the 26 auctions with participation, 14 were undersubscribed. Only three consecutive Schedule 1 auctions in the fall of 2008—following Lehman's bankruptcy—saw a considerable hike in the stop-out rate. These were on September 18, October 2, and October 16, with stop-out rates of 151 bps, 42 bps, and 46 bps, respectively. Overall, 15 auctions awarded loans at a stop-out rate of 10 bps (the minimum bid) and eight awarded loans within 6 bps of the minimum bid.

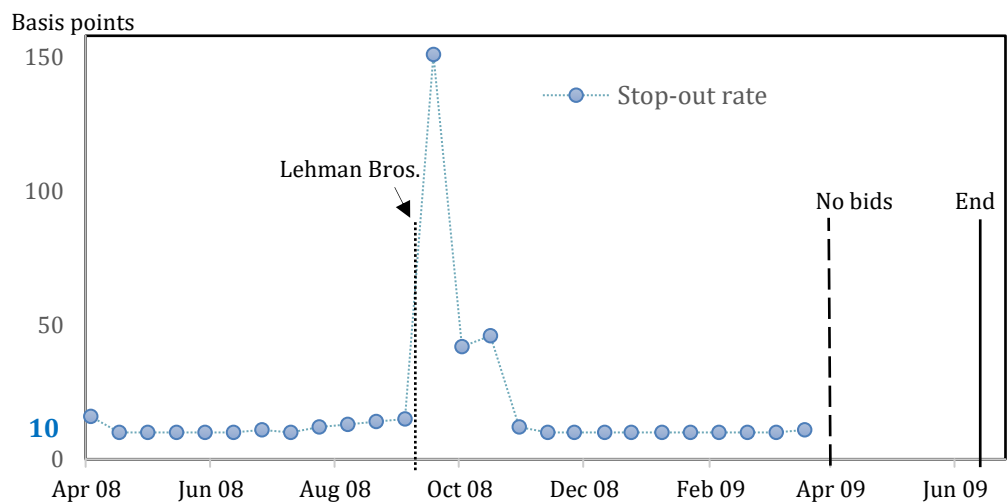
Figure 5: Utilization of TSLF Schedule 1



Note: Each bar represents a Schedule 1 auction. In total, there were 33 between April 3, 2008, and June 25, 2009.

Source: Author’s analysis of FRBNY data.

Figure 6: Stop-Out Rates for TSLF Schedule 1

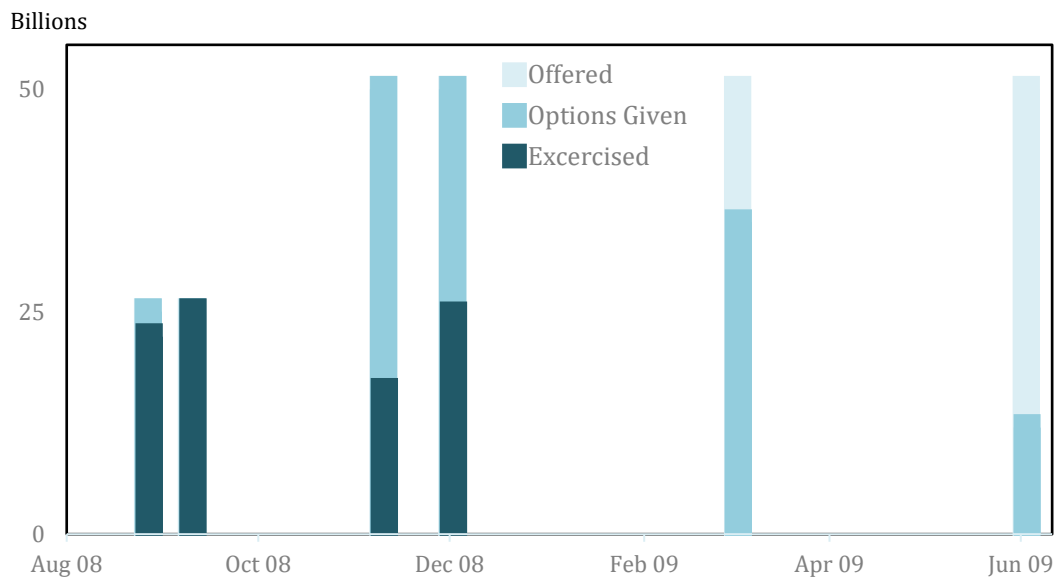


Note: The minimum bid rate was 10 bps. No bids after March 19, 2009. The last auction was held on June 25, 2009.

Source: Author’s analysis of FRBNY data.

The FRBNY held six TOP auctions between August 27, 2008, and June 3, 2009. The initial two auctions of August 27 and September 10 offered \$25 billion of US Treasury securities with a fixed rate of 25 bps. Both of these auctions were fully subscribed, and dealers were awarded the options to borrow US Treasury securities on September 24, at stop-out rates of 2 bps and 3 bps, respectively. The remaining four TOP auctions offered \$50 billion in options. These were held in the end of 2008, on November 10, and December 2, and in 2009, on March 3, and June 3. The two auctions in 2009 were undersubscribed, and the options awarded were not exercised. The TOP ended in July 2009, in view of low demand and improved market conditions.

Figure 7: Utilization of TOP



Note: Each bar represents a TOP auction. In total, there were six between August 27, 2008, and June 3, 2009. Of notice was that Cantor Fitzgerald & Co. did not exercise a \$2 billion option received in the first auction.

Source: Author's analysis of FRBNY data.

Figure 8: Stop-Out Rates for TOP Options

Date	Aug 27, 2008	Sep 10, 2008	Nov 10, 2008	Dec 2, 2008	Mar 3, 2009	Jun 3, 2009
Basis Points	2	3	2	5	1	1

Note: The FRBNY auctioned options to loan US Treasury securities at a fixed rate, about two to three weeks after the auction date.

Source: FRBNY data.

Figure 9: Fixed Rates of US Treasury Securities under TOP

Date	Aug 27, 2008	Sep 10, 2008	Nov 10, 2008	Dec 2, 2008	Mar 3, 2009	Jun 3, 2009
Basis Points	25	25	50	50	25	25

Source: FRBNY data.

II. Key Design Decisions

1. The Federal Reserve invoked the emergency powers under FRA Section 13(3) to establish the Term Securities Lending Facility.

In March 2008, the 2007–09 financial crisis reached a critical stage. Amid falling house prices and downgrades of mortgage-related securities, financial markets became severely disrupted. In response, the Federal Reserve Board, in order to act as lender of last resort to primary dealers, invoked the emergency powers of Section 13(3) of the Federal Reserve Act, last used during the Great Depression in the 1930s. Under Section 13(3), the Board could decide to lend to nondepository institutions in “unusual and exigent circumstances.” The approval of a majority vote of the FOMC was also required because the action would affect open market operations. On March 10, 2008, with a unanimous vote of the five sitting Board Governors of the Federal Reserve System and a 9–0 vote of the FOMC, the TSLF was approved (Bernanke 2015).

2. The TSLF was administered by the Federal Reserve Bank of New York.

The Fed, primarily through the FRBNY, crafted the TSLF. In the open market operations, the FRBNY trades US government securities with designated primary dealers (securities firms) on a regular basis. The TSLF took advantage of existing infrastructure and some similarities in design with the OMOs (Federal Reserve 2009; Geithner 2014).

3. The FRBNY was authorized to lend up to \$200 billion in US Treasury securities.

In view of the great demand in the markets for US Treasury securities as a “safe haven,” the FRBNY lent highly liquid US Treasury securities such as Treasury bills, notes, bonds, and inflation-indexed securities, at a 28-day term—much longer than the overnight term—and against a broader range of collateral. Initially, the list of eligible collateral included collateral that was illiquid at the time, such as triple-A private-label residential mortgage-backed securities and commercial MBS, as well as agency collateralized mortgage obligations. The FRBNY drew the US Treasury securities from the System Open Market Account. After financial markets improved in June 2009, the Fed reduced its size to \$75 billion. The TSLF ended in February 2010 (Federal Reserve 2009; Federal Reserve 2010).

4. The TSLF lent highly liquid US Treasury securities, not cash.

While other Fed lending facilities involved cash, the TSLF involved only securities. Through the TSLF, primary dealers settled repurchase agreements in which they received highly liquid US Treasury securities in exchange for less liquid securities.

Officials from the FRBNY have stated that the TSLF, as a securities-for-securities lending program, did not affect the supply of bank reserves and that “the benefit was that the Fed was not adding more cash to the economy and therefore did not have to take offsetting actions to manage the fed funds rate. This was a key difference in TSLF from the Single-Tranche repo program and from the Primary Dealer Credit Facility, and meant it could be scaled up or down more quickly” (Logan, Nelson, and Parkinson 2018).

5. TSLF loans were recourse.

The loans were recourse. That is, in case of default, the Fed could go after the primary dealer’s assets to claim for the difference between the liquidation of the pledged collateral and the value of the loan (GAO 2011).

6. Participation was limited to the 20 primary dealers of the FRB-NY.

Only the 20 primary dealers of the FRBNY, including the five largest US investment banks and the US securities arms of major foreign financial institutions, were eligible to participate in the TSLF (Fed 2008).

7. The FRBNY awarded loans through weekly auctions.

With an auction mechanism, the Fed intended to encourage broad participation and avoid any stigmatization in using this facility. The upper limits for primary dealers on the share of the auction that they could get guaranteed that multiple of them would be awarded TSLF loans. The TSLF consisted of weekly auctions that lasted for 30 minutes. The day before an auction, the Fed announced the amount and securities offered. Participation was voluntary and bids represented the fee dealers intended to pay to loan the offered US Treasury securities. Auction results were communicated shortly thereafter (Federal Reserve 2008b).

8. The auctions were classified into two categories, including a Schedule 2 for relatively illiquid assets.

Auctions were classified into two categories: Schedule 1 and Schedule 2. While Schedule 1 auctions accepted all collateral eligible in FRBNY open market operations, Schedule 2 auctions accepted a much broader and less liquid range of collateral and required a higher minimum bid. The two different schedules were intended “to better calibrate the interest rate on TSLF loans to the level of risk associated with the collateral” (GAO 2011).

Schedule 1 eligible collateral included: Treasury securities, agency debt securities, and agency MBS.; whereas Schedule 2 collateral included: all Schedule 1 eligible collateral, plus triple-A private-label residential MBS and commercial MBS, and agency CMOs (FRBNY 2008a). On April 29, 2008, Schedule 2 collateral was expanded to include triple-A asset-backed securities (Federal Reserve 2008b). And on September 14, 2008, a day prior to the collapse of Lehman Brothers, the Board and the FOMC broadened the list to include all investment-grade debt securities (BBB- or higher) (Federal Reserve 2008c).

9. The TSLF weekly auctions alternated between Schedule 1 and Schedule 2 collateral and the timing was revised shortly before Lehman Brothers failed.

TSLF auctions started on March 27, 2008, and ran weekly, alternating between Schedule 1 and Schedule 2, until September 11, 2008. In these first months of operation, the Fed offered \$175 billion of the available \$200 billion: \$50 billion for Schedule 1 auctions and \$125 billion for Schedule 2 auctions. Schedule 1 auctions offered \$25 billion each, while Schedule 2 auctions alternated offering amounts between \$50 billion and \$75 billion.

On September 14, 2008, one day before Lehman filed for bankruptcy, the Fed revised the timing of Schedule 2 auctions to run weekly, rather than biweekly. It also increased the size from \$125 billion to \$150 billion per month. From September 17, 2008, until April 15, 2009, all Schedule 2 weekly auctions offered \$37.5 billion (only three \$35 billion). As financial conditions improved in 2009, the size and timing of Schedule 2 auctions was decreased, and they eventually ended on January 7, 2010.

10. Margins (haircuts) were imposed according to the type of collateral.

The FRBNY determined margin requirements. These did not change throughout the crisis. Haircuts aimed to reduce the Fed's risk in case of bankruptcy of a primary dealer (Federal Reserve 2009). Additionally, collateral on review for downgrade was not accepted. For securities posted as collateral that were on review for downgrade, the Fed could demand that they be replaced with other eligible securities (FRBNY 2009).

11. Two clearing banks provided the custody and valuation of collateral.

The FRBNY relied on the clearing services of the existing clearing banks in the triparty repo market, JPMorgan Chase and Bank of New York Mellon. They functioned as intermediaries and provided the services of custody and valuation of collateral. Their services came at no charge to the Federal Reserve or borrowers.

The clearing banks conducted daily revaluations of collateral to make sure the specified margin was applied. The US Treasury securities awarded remained in the clearing banks, but dealers could use them to settle repo contracts. In case the value of the collateral decreased, the FRBNY could ask for substitutions. On the other hand, primary dealers could also substitute collateral for other eligible collateral if needed. To value the pledged collateral, clearing banks used the lowest price available in their valuation systems (Federal Reserve 2009).

12. Minimum bids were set at levels considered low during the crisis but high during normal times.

The minimum bid for Schedule 1 auctions was 10 bps and for Schedule 2 was higher at 25 bps. There was no cost to place a bid (FRBNY 2008). Many of the Fed's lending programs during the crisis were intended to be self-liquidating as markets improved. Minimum bid

rates and collateral requirements were set to be attractive when markets were disrupted but unattractive when markets functioned well (Kohn 2008).

13. Primary dealers had a limit on borrowing.

Primary dealers could place up to two bids, with a minimum amount of \$10 million and not higher than 20% of the total offered amount. In case of two bids, they could only be in increments of \$10 million. Primary dealers could be awarded a maximum of only 20% of the total offering amount, regardless if the auction was undersubscribed. The Fed set limits “to ensure that the lending [was] distributed across multiple institutions” (Logan 2009). The FRBNY held the right to refuse any bid at its own discretion (FRBNY 2008a). Awards were determined on a pro rata basis, and dealers were not allowed to terminate a loan early (Federal Reserve 2008).

14. Loans were awarded at a uniform fee, based on the lowest accepted bid—the “stop-out rate.”

At the end of the auction, the uniform fee was set based on the stop-out rate, which was the lowest accepted bid (GAO 2011).

15. The TSLF Options Program was an extension to the TSLF.

The TOP was approved on July 24, 2008, and announced on July 30, 2008. It was an extension to the TSLF and required administrative changes pursuant to Schedule 2 auctions and not another FRA Section 13(3) authorization. The TOP intended to provide primary dealers some relief during periods of “heightened collateral market pressure, such as quarter-end dates.”

The new program authorized the FRBNY to offer “options” to loan US Treasury securities for a seven-day term, two to three weeks after the auction date and at a fixed rate. In some way, the TOP was a pre-auction that gave primary dealers the guarantee that they could have access to liquid assets at a later date, without the commitment to exercise the options. The TOP was limited to \$50 billion in US Treasury securities, on top of the \$200 billion of Schedule 1 and Schedule 2 programs. The minimum bid rate was 1 bp, and all Schedule 2 collateral was eligible. The day before each TOP auction, the FRBNY announced the terms and conditions, including the fixed rate of the loans. The TOP held its first auction on August 27, 2008 (OIG 2010.).

16. While the TSLF was operational, the identity of the borrowers was undisclosed.

While the TSLF was in operation, the Fed published information on awards, stop-out rates, and bid-to-cover ratios, but it did not disclose the identity of the participant primary dealers or the bid propositions.

In July 2010, the US Congress passed Dodd-Frank. The measures included to promote transparency required the Fed to disclose certain information on their emergency lending programs. In regard to the TSLF, on December 1, 2010, the Fed disclosed detailed information about borrowers between December 1, 2007, and July 21, 2010. The information included the identity of the borrowers, dates, type and amounts of financial assistance provided, the interest charged, and pledged collateral.

III. Evaluation

The nature of the TSLF as one of many temporary emergency lending facilities makes it hard to evaluate. However, it would be useful to consider at least three components: 1) the effects the announcement had on financial markets, 2) the effectiveness of the TSLF in providing relief to primary dealers or the repo market, and 3) the auction design and legislative tools.

The effect of the TSLF announcement is unclear. While Fed Chairman Bernanke (2015) was of the opinion that “market participants applauded,” US Treasury Secretary Paulson (2010) thought that “... the opposite happened. It was an indication of the markets’ jitters that some took the move as a confirmation of their worst fears: things must be very serious indeed for the Fed to take such unprecedented action.”

Fleming, Hrung, and Keane (2010) and Hrung and Seligman (2011) have argued that the TSLF contributed in the reduction of stress on repo markets. Acharya et al. (2017), using proprietary data, concluded that primary dealers that possessed less liquid collateral, lower equity returns, and greater leverage prior to the crisis were more inclined to borrow and at higher bidding rates.

The Fed’s emergency lending facilities under Section 13(3), such as the TSLF, raised concerns among the public for the lack of transparency on the valuation of pledged collateral and the undisclosed identity of the borrowers. In May 2008, a Bloomberg News reporter requested that the Fed Board, under the Freedom of Information Act (FOIA) and via email, provide detailed information with respect to securities posted as collateral for multiple emergency lending facilities, such as the TSLF. In the court case *Bloomberg L.P. v. Board of Governors of the Federal Reserve System*, Fed officials cited adverse effects to the disclosure of the detailed information such as stigma and the potential reluctance of borrowers to use the facilities in the future. In addition, they added that the disclosure of “highly sensitive” information could cause “substantial competitive harm.” Furthermore, the “public disclosure of information regarding specific securities pledges as collateral for individual TSLF loans would significantly harm the Government’s monetary functions or commercial interests” (Logan 2009; Madigan 2009). Ultimately, in 2011, the court ruled that the Fed was required to release some of the information solicited.

In July 2010, the Congress passed Dodd-Frank. Measures to promote transparency required the Fed to disclose information on their emergency lending programs between 2007 and 2010. In regard to the TSLF, on December 1, 2010, the Fed disclosed detailed information about borrowers between December 1, 2007, and July 21, 2010. The information included the identity of the borrowers, dates, type and amounts of financial assistance provided, the interest charged, and pledged collateral.¹⁴ It appears that the TSLF as a temporary emergency lending facility had an implicit agreement that specific information on participants was going to remain undisclosed. However, the outcome of the Bloomberg FOIA case and the Dodd-Frank act required the Fed to be more transparent. These outcomes have set a precedent that could change the perceptions of the Fed and primary dealers in designing a similar facility in the future.

IV. References

Acharya, Viral V., Michael J. Fleming, Warren B. Hrung, and Asani Sarkar. 2017. “Dealer Financial Conditions and Lender-of-Last-Resort Facilities.” *Journal of Financial Economics*

¹⁴ For the Fed’s disclosure, see http://www.federalreserve.gov/newsevents/reform_transaction.htm.

123, no. 1: 81–107.
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Dealer_Fin_Conds_and_LOL_R_Facilities.pdf

Bernanke, Ben S. 2015. *The Courage to Act: A Memoir of a Crisis and Its Aftermath*. New York: W.W. Norton.
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/bk%20One%20Pagers.pdf>.

Carlson, M. & Macchiavelli, M. (2018). "Emergency Collateral Upgrades," Finance and Economics Discussion Series 2018-078. Washington: Board of Governors of the Federal Reserve System.
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/2018078pap.pdf>.

Board of Governors of the Federal Reserve System (Federal Reserve). 2008a. "FOMC Statement: Federal Reserve and Other Central Banks Announce Specific Measures Designed to Address Liquidity Pressures in Funding Markets." Press Release, March 11, 2008.
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FOMC_Fed_central_banks_announce_measures_to_address_liquidity_pressures.pdf.

Board of Governors of the Federal Reserve System (Federal Reserve). 2008b. Minutes of the Federal Open Market Committee, April 29–30, 2008.
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/fomcminutes20080430.pdf>.

Board of Governors of the Federal Reserve System (Federal Reserve). 2008c. "Federal Reserve Board Announces Several Initiatives to Provide Additional Support to Financial Markets, Including Enhancements to Its Existing Liquidity Facilities." Press Release, September 14, 2008.
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRB_PR_09-14-2008.pdf.

Board of Governors of the Federal Reserve System (Federal Reserve). 2009. "Report Pursuant to Section 129 of the Emergency Economic Stabilization Act of 2008: Term Securities Lending Facility."
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/129tslf.pdf>.

Board of Governors of the Federal Reserve System (Federal Reserve). 2010. "Federal Reserve Releases Detailed Information about Transactions Conducted to Stabilize Markets during the Recent Financial Crisis." Press Release, December 1, 2010.
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20releases%20detailed%20information%20about%20transactions%20conducted%20to%20stabilize%20markets%20during%20the%20recent%20financial%20crisis.pdf>.

Board of Governors of the Federal Reserve System (Federal Reserve). 2018. "Term Securities Lending Facility (TSLF) and TSLF Options Program (TOP)." Accessed December 12, 2018.
[https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/The%20Fed%20-%20Term%20Securities%20Lending%20Facility%20\(TSLF\)%20and%20TSLF%20Options%20Program%20\(TOP\).pdf](https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/The%20Fed%20-%20Term%20Securities%20Lending%20Facility%20(TSLF)%20and%20TSLF%20Options%20Program%20(TOP).pdf).

Board of Governors of the Federal Reserve System, Office of Inspector General (OIG). 2010. *The Federal Reserve's Section 13(3) Lending Facilities to Support Overall Market Liquidity: Function, Status, and Risk Management*. Washington, DC: Board of Governors of the Federal

Reserve System, Office of Inspector General, November 16, 2010. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRS_Lending_Facilities_Report_final-11-23-10_web.pdf.

Federal Reserve Bank of New York (FRBNY). 2008a. "New York Fed Announces Modifications to Terms and Conditions of Term Securities Lending Facility." Press Release, March 20, 2008. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRBNY_Announces_Mods_to_Terms_Conds_TSLF.pdf.

Federal Reserve Bank of New York (FRBNY). 2009. "Term Securities Lending Facility: Frequently Asked Questions." https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/TSLF_FAQ_09-03-24.pdf.

Federal Reserve Bank of New York (FRBNY). 2018a. "Term Securities Lending Facility." <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Term%20Securities%20Lending%20Facility%20-%20FEDERAL%20RESERVE%20BANK%20of%20NEW%20YORK.pdf>.

Federal Reserve Bank of New York (FRBNY). 2018b. "Term Securities Lending Facility: Historical Search." <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Term%20Securities%20Lending%20Facility%20-%20Federal%20Reserve%20Bank%20of%20New%20York.pdf>.

Federal Reserve Bank of St. Louis (FRBSL). "Federal Reserve Bank of St. Louis' Financial Crisis Timeline." https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Bank%20of%20St.%20Louis'%20Financial%20Crisis%20Timeline%20Timeline%20-%20FRASER%20_%20FRASER%20_%20St.%20Louis%20Fed.pdf.

Fleming, Michael J., Warren B. Hrungr, and Frank M. Keane. 2010. "Repo Market Effects of the Term Securities Lending Facility." *American Economic Review* 100, no. 2 (May): 591–96. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Repo_Market_Effects_of_TSLF_0.pdf.

Geithner, Timothy F. 2014. *Stress Test: Reflections on Financial Crises*. New York: Crown Publishers. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/book_stress_test.pdf.

Gorton, Gary, and Andrew Metrick. 2010. "Haircuts." Yale ICF Working Paper, no. 09–15, May http://depot.som.yale.edu/icf/papers/fileuploads/2395/original/2010_ICF_WPS_Haircuts-Metrick_Gorton.pdf.

Gorton, Gary, and Andrew Metrick. 2012. "Securitized Banking and the Run on Repo." *Journal of Financial Economics* 104, no. 3: 425–51. <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Gorton,%20Gary%20B%20&%20Andrew%20Metrick%20Securitized%20Banking%20&%20Run%20on%20Repo%202012.pdf>

Hrungr, Warren B., and Jason S. Seligman. 2011. "Responses to the Financial Crisis, Treasury Debt, and the Impact on Short-Term Money Markets." *Federal Reserve Bank of New York Staff Reports*, no. 481 (January). <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/sr481.pdf>.

Kohn, Donald L. 2008. "Money Markets and Financial Stability." Speech at the Federal Reserve Bank of New York and Columbia Business School Conference on the Role of Money Markets, New York, May 29, 2008. <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Money%20Markets%20and%20Financial%20Stability%20-%20Federal%20Reserve%20Board.pdf>.

Logan, Lorie, William Nelson, and Patrick Parkinson. 2018. "Novel Lender of Last Resort Programs." Conference on Responding to the Global Financial Crisis: What We Did and Why We Did It, Brookings Institution, Washington, DC, September 11–12. <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/02%20Novel%20LOLR%20Prelim%20Disc%20Draft%202018.09.11.pdf>.

Madigan, Brian F. 2009. Declaration of Brian F. Madigan, March 1, 2009. *Bloomberg L.P. v. Board of Governors of the Federal Reserve System*, 601 F.3d 143 (2d Cir. 2010). <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/0.pdf>.

Paulson, Henry M, Jr. 2010. *On the Brink: Inside the Race to Stop the Collapse of the Global Financial System*. New York: Business Plus, Hachette Book Group. <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Paulson%20YPFS%20Resource%20Library%20One%20Pagars.pdf>.

US Government Accountability Office (GAO). 2011. *Opportunities Exist to Strengthen Policies and Processes for Managing Emergency Assistance*. GAO-11-696. Washington, DC, July 2011. https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/GAO_Report_07-2011.pdf.

V. Key Program Documents

Summary of Program

Report Pursuant to Section 129 of the Emergency Economic Stabilization Act of 2008: Term Securities Lending Facility (Federal Reserve 2009) – *The US Federal Reserve filed the TSLF report to the US Congress pursuant to section 129 of the Emergency Economic Stabilization Act of 2008. The report provided updates concerning the TSLF which was established by the Board under section 13(3) of the Federal Reserve Act.* <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/129tslf.pdf>.

The Federal Reserve's Section 13(3) Lending Facilities to Support Overall Market Liquidity: Function, Status, and Risk Management, November 16, 2010 (Federal Reserve 2010) – *The Office of Inspector General presented a report on the Federal Reserve's Section 13(3) Lending Facilities implemented during the 2007-09 financial crisis. This report provided an independent review of the six lending facilities' functions, status, and risks.* https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRS_Lending_Facilities_Report_final-11-23-10_web.pdf.

Term Securities Lending Facility: Frequently Asked Questions (Federal Reserve Bank of New York 2008) – *The TSLF: Frequently Asked Questions addresses operational questions about the Term Securities Lending Facility (TSLF).* https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/TSLF_FAQ_09-03-24.pdf.

Term Securities Lending Facility: Program Terms and Conditions (Federal Reserve Bank of New York 2008) – *The TSLF: Program Terms and Conditions provides guidance on the TSLF's terms and conditions such as auctions dates, eligibility securities, fees, and minimum bids.*

https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/TSLF_Terms_Conds_09-03-24.pdf.

Legal/Regulatory Guidance

Federal Reserve Act Section 13. Powers of Federal Reserve Banks (Federal Reserve 2018) – *The Federal Reserve Act, Section 13 stipulates the powers of Federal Reserve Banks. In particular, the Federal Reserve may invoke Section 13(3) in “unusual and exigent circumstances” for discounts for individuals, partnerships, and corporations.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRA_Section_13_Sept_19_2008_0.pdf.

Implementing The Dodd-Frank Act: The Federal Reserve Board’s Role (Federal Reserve 2018) – *The US Federal Reserve Board, responsible for issuing rules under the Dodd-Frank Wall Street Reform and Consumer Protection Act, provided a list of initiatives completed, along with proposals the Board planned through 2013.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/The%20Fed%20-%20Dodd-Frank%20Act_%20Statutory%20Dates%20For%20Actions.pdf.

Dodd-Frank Wall Street Reform and Consumer Protection Act (US Congress (2010) – *Public Law 111-203—July 21, 2010. The Act enacted in the U.S. Congress stated “to promote the financial stability of the United States by improving accountability and transparency in the financial system, to end “too big to fail”, to protect the American taxpayer by ending bailouts, to protect consumers from abusive financial services practices, and for other purposes.”*
<https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/PUBLIC%20LAW%20111%20203%2007-21-2010.pdf>.

Press Releases/Announcements

FOMC Statement: Federal Reserve and Other Central Banks Announce Specific Measures Designed to Address Liquidity Pressures in Funding Markets (March 11, 2008) – *The Federal Reserve, in conjunction with the Bank of Canada, the Bank of England, the European Central Bank, and the Swiss National Bank announced measures to address liquidity, which included the establishment of the TSLF.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FOMC_Fed_central_banks_announce_measures_to_address_liquidity_pressures.pdf.

Federal Reserve, European Central Bank, and Swiss National Bank Announce an Expansion of Liquidity Measures (May 2, 2008) – *The Federal Reserve expanded the collateral that primary dealers could be pledged for the TSLF Schedule 2 auctions to include AAA/Aaa-rated asset-backed securities.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Fed_ECB_SNB_announce_expansion_liquidity_measures.pdf.

Federal Reserve Announces Steps to Enhance the Effectiveness of Its Existing Liquidity Facilities (July 30, 2008) – *The Federal Reserve expanded the TSLF to include a TSLF Options Program for \$50 billion. It offered primary dealers through auctions the option to borrow U.S. Treasury securities for a 7-day term, two to three weeks after the auction date and at a fixed rate. The Federal Reserve intended to “offer such options for exercise in advance of periods that are typically characterized by elevated stress in financial markets, such as quarter ends.”*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Fed_announces_steps_enhance_effectiveness_existing_liquidity_facilities.pdf.

Federal Reserve Board Announces Several Initiatives to Provide Additional Support to Financial Markets, Including Enhancements to Its Existing Liquidity Facilities (September 14, 2008). – *The Federal Reserve Board announced multiple initiatives to support financial markets, including enhancements to existing liquidity facilities. The collateral for the TSLF was expanded for Schedule 2 auctions to include all investment-grade debt securities.* https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/FRB_PR_09-14-2008.pdf.

Federal Reserve Announces the Extension of Three Liquidity Facilities through April 30, 2009 (December 2, 2008) – *Due to continuing strains in financial markets, the Federal Reserve extended through April 30, 2009, three liquidity facilities: the PDCF, the AMLF, and the TSLF.* <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20announces%20the%20extension%20of%20three%20liquidity%20facilities%20through%20April%2030,%202009.pdf>.

Federal Reserve Announces Extension through October 30, 2009, of Its Existing Liquidity Programs That Were Scheduled to Expire on April 30, 2009 (February 3, 2009) – *The Federal Reserve extended through October 30, 2009 liquidity programs scheduled to expire on April 30, 2009 that include the AMLF, the CPFF, the MMIFF, the PDCF, and the TSLF.* <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20announces%20extension%20through%20October%2030,%202009,%20of%20its%20existing%20liquidity%20programs%20that%20were%20scheduled%20to%20expire%20on%20April%2030,%202009.pdf>.

Federal Reserve Announces Extensions of and Modifications to a Number of Its Liquidity Programs (June 25, 2009) – *The Federal Reserve, in view of very weak demand for TSLF auctions in previous months, suspended TSLF Schedule 1 and TSLF Options Program auctions. It reduced the frequency and size of TSLF Schedule 2 auctions.* <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20announces%20extensions%20of%20and%20modifications%20to%20a%20number%20of%20its%20liquidity%20programs.pdf>.

Federal Reserve Announces Term Auction Facility (TAF) and Term Securities Lending Facility (TSLF) Schedules through January 2010 (September 24, 2009) – *The Federal Reserve extended TAF and TSLF operations through January 2010. In light of continuing improvements in financing markets, the amounts offered in TSLF auctions were reduced.* [https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20announces%20Term%20Auction%20Facility%20\(TAF\)%20and%20Term%20Securities%20Lending%20Facility%20\(TSLF\)%20schedules%20through%20January%202010.pdf](https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20announces%20Term%20Auction%20Facility%20(TAF)%20and%20Term%20Securities%20Lending%20Facility%20(TSLF)%20schedules%20through%20January%202010.pdf).

Federal Reserve Releases Detailed Information about Transactions Conducted to Stabilize Markets during the Recent Financial Crisis (December 1, 2010) – *As required by the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010, the Federal Reserve posted transaction-level details from December 1, 2007, to July 21, 2010, on multiple lending facilities employed during the 2007-09 financial crisis, including the TSLF.* <https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Federal%20Reserve%20Board%20-%20Federal%20Reserve%20releases%20detailed%20information%20about%20transactions%20conducted%20to%20stabilize%20markets%20during%20the%20recent%20financial%20crisis.pdf>.

Term Securities Lending Facility: Announcements (Accessed December 12, 2018) – *The Federal Reserve Bank of New York provided a timeline of important announcements regarding the TSLF.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/Term%20Securities%20Lending%20Facility_%20Announcements%20-%20FEDERAL%20RESERVE%20BANK%20of%20NEW%20YORK.pdf.

Reports/Assessments

Opportunities Exist to Strengthen Policies and Processes for Managing Emergency Assistance, July 2011 (US Government Accountability Office 2011) – *As directed by the Dodd-Frank Wall Street Reform and Consumer Protection Act, the US Government Accountability Office conducted an audit of the emergency loan programs and other assistance authorized by the Federal Reserve Board during the 2007-09 financial crisis.*
https://ypfsresourcelibrary.blob.core.windows.net/fcic/YPFS/GAO_Report_07-2011.pdf.

Copyright 2015, 2016, 2020 © Yale University. All rights reserved. To order copies of this material or to receive permission to reprint any or all of this document, please contact the Yale Program for Financial Stability at ypfs@yale.edu.

Appendix A: TSLF Timeline

Date	Event
Mar/11/2008	TSLF announced. Auctions of highly liquid US Treasury securities to primary dealers of up to \$200 billion per month.
Mar/20/2008	TSLF details and eligible collateral for each type of auction announced: Schedule 1: \$50 billion per month. Auctions of \$25 billion each. Treasury securities, agency debt securities, and agency mortgage-backed securities (MBS). Schedule 2: \$125 billion per month. Alternating auctions of \$50 billion and \$75 billion. All Schedule 1 eligible collateral, triple-A private-label residential MBS and commercial MBS, and agency collateralized mortgage obligations.
Mar/27/2008	First TSLF auction: Schedule 2 for \$75 billion.
Apr/29/2008	Schedule 2: eligible collateral expanded to include triple-A asset-backed securities (ABS).
Jul/24/2008	TSLF Options Program (TOP) of \$50 billion approved. TSLF extended until January 30, 2009.
Aug/27/2008	First TOP auction.
Sep/14/2008	Schedule 2: size increased from \$125 billion to \$150 billion per month. Auctions to run weekly for \$37.5 billion (most cases). Eligible collateral expanded to include all investment-grade securities (BBB- or higher).
Sep/15/2008	Lehman Brothers collapsed. The largest bankruptcy in US history.
Oct/01/2008	TSLF reaches its peak usage of \$236 billion.
Dec/02/2008	TSLF extended until April 30, 2009.
Jan/27/2009	TSLF extended through October 30, 2009.
Jun/03/2009	Last TOP auction.
Jun/23/2009	TSLF extended until February 1, 2010, but only for Schedule 2 auctions. Schedule 2: size decreased to \$75 billion per month. Auctions to run once a month.
Jun/25/2009	Last Schedule 1 auction.
Jul/16/2009	Last Schedule 2 loans given.
Sep/22/2009	Schedule 2: size decreased to \$50 billion for October and \$25 billion thereafter.
Jan/07/2010	Last Schedule 2 auction.
Feb/01/2010	TSLF ended.